Methodologies for the correction of "Garbage Codes" in mortality records and their applicability in microsimulation models: Protocol for a scoping review

Abstract

Objective: To systematically map and synthesize the scientific evidence on the quantitative methodologies used for the identification, cleaning and redistribution of "garbage codes" in population-based mortality registries, evaluating their technical applicability for the parameterization of microsimulation models in health.

Inclusion criteria: This review will consider studies utilizing population-based mortality records or vital statistics (Population). Eligible literature must describe or apply traditional statistical methods, multiple imputation, Bayesian models, record linkage, or machine learning algorithms to correct ill-defined causes of death and "garbage codes" (Concept). The context includes health information systems characterized by scarce or heterogeneous data, specifically where the feasibility of disaggregating results to the individual level to parameterize stochastic microsimulation models is analyzed (Context).

Methods: This scoping review will follow the JBI methodology and the PRISMA Extension for Scoping Reviews (PRISMA-ScR). Comprehensive searches will be performed across PubMed, Scopus, Web of Science, LILACS, and SciELO, supplemented by grey literature from the WHO, PAHO, and IHME. To ensure reproducibility, the workflow will be executed within the R statistical environment. We will utilize litsearchr for search syntax optimization, synthesisr for deduplication, and revtools for blinded screening supported by latent Dirichlet allocation topic clustering. Discrepancies will be resolved by discussion or a third reviewer. Following paired data extraction, findings will be synthesized narratively and visualized using ggplot2 matrix heatmaps and igraph flow networks.

Keywords: health information systems; causes of death; vital statistics; data quality; machine learning algorithms; proportional redistribution; computer simulation.

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1 INTRODUCTION

Civil registration and vital statistics systems serve as the cornerstone of demographic and medium- to long-term health intelligence[1,2], underpinning policy planning, health technology assessment[3–5], and the quantification of the true impact of health crises[6,7]. However, despite decades of global advocacy, data quality progress remains sluggish[1,2,8], and the analytical utility of these databases is frequently undermined by persistent medical certification errors[7,9–11]. The primary challenge is the assignment of deaths to "garbage codes" (GCs), defined as International Classification of Diseases codes that represent ambiguous, non-specific, or biologically implausible underlying causes of death[12–14]. Because raw mortality data frequently lack diagnostic precision, the application of indirect statistical correction remains a crucial public health priority[15], particularly given the escalating burden of non-communicable diseases[16] and cancers in data-scarce regions[17,18]. Furthermore, because clinical interventions and medical training alone cannot fully eliminate these diagnostic errors at the source[7,9], robust post-hoc redistribution methods are indispensable[13,19].

To mitigate these systematic biases, a diverse array of correction methodologies has evolved over the past two decades[6,19–22]. Modern redistribution strategies are typically classified into four distinct mathematical approaches: multiple cause analysis, negative correlation, impairment reallocation, and proportional redistribution[11]. Beyond these foundational methods, researchers have increasingly utilized individual-level multiple cause of death data to capture hidden etiologies[23–25]. This led to the development of advanced computational techniques, including clinical data linkage algorithms[26] and coarsened exact matching (CEM, a method that groups continuous variables into discrete strata to pare mortality records precisely without parametric assumptions)[27,28].

Recent innovations also include subnational regression models[29–31], four-step probabilistic algorithms[32], the application of pathophysiological redistribution packages[33], and the integration of international rules with exact matching[34]. Moreover, spatial and machine learning approaches have emerged to address complex data patterns: spatial scan statistics and Local Moran's I autocorrelation adjust for spurious geographical clusters[35,36], while random forest algorithms and decision trees map spatial inconsistencies and hierarchical clinical interactions[37,38].

Before applying these algorithms, baseline data quality is now objectively assessed using standardized metrics like the Vital Statistics Performance Index[8,39] and automated diagnostic tools like ANACONDA[10,40]. However, despite this methodological proliferation, current literature exhibits a critical translational gap[41]. Specifically, there is a lack of evidence evaluating how macro-level redistribution assumptions can be rigorously transferred to individual-level simulation environments[42,43].

Despite the proliferation of macro-level redistribution frameworks, current literature exhibits a critical translational gap[32,41]. Aggregated population-level corrections assume that misclassified or ill-defined causes of death are randomly distributed across demographic strata[11,31]. However, stochastic microsimulations and agent-based models require highly disaggregated baseline mortality risks, as structural diagnostic errors disproportionately mask true epidemiological burdens in vulnerable populations and settings with high social deprivation[44–46]. Translating uncorrected macro-level proportions directly into individual-level simulation environments risks compounding classification errors and propagating systematic bias into microdata risk matrices[42,43]. Therefore, this scoping review aims to map the state of the art, evaluate the technical feasibility of adapting macro-level correction algorithms for individual-level microsimulation parameterization, and establish a conceptual framework to guide future public health mathematical modeling.

1.1 Research questions

1. What analytical methodologies (statistical, probabilistic, deterministic, linkage-based or machine learning) have been described in the literature for the correction and redistribution of "garbage codes" in mortality records?
2. How are they classified and what types of "garbage codes" (poorly defined causes of ICD chapters, intermediate or terminal causes) are the main objective of each identified methodology?
3. What is the documented level of applicability or compatibility of these correction methods to structure databases aimed at the development, calibration and validation of microsimulation models in health?
4. What are the main methodological challenges, limitations and advantages reported when transferring or adapting analytical correction algorithms from the population (macro) level to the individual level (micro)?

2 ELIGIBILITY CRITERIA

2.1 Population

This component defines eligible data records, populations, and geographic or demographic coverages.

- **Inclusion criteria:**

  - **Data type:** Studies utilizing population-based mortality databases, vital statistics systems, official national or subnational health statistics registers, or civil registration and vital statistics (CRVS) systems.
  - **Demography:** Records tracking general human mortality without restrictions on age groups (including neonatal, infant, adult, or geriatric mortality) or sex and gender.

- **Exclusion criteria:**
  - Studies restricted to hyper-specific hospital specimens, controlled trials, or clinical registries (e.g., in-hospital mortality tracking for acute coronary syndromes or percutaneous coronary interventions) that do not utilize—or aim to correct—broader population-based mortality records or epidemiological vital statistics systems.

2.2 Concept

This component delineates the core methodologies, algorithms, and analytical tools evaluated in this review.

- **Inclusion criteria:**
  - Studies describing, evaluating, validating, or implementing quantitative methods to identify, clean, reclassify, or redistribute "garbage codes" (levels 1 to 4) or ill-defined causes of death (as classified under ICD-10/ICD-11 Chapter XVIII, or ICD-9 equivalents).
  - Methodologies within the following analytical categories are eligible:
    * **Traditional statistics and mathematical demography:** Proportional redistribution (simple or stratified by age and sex), fractional algorithms, and indirect demographic adjustment methods[21].
    * **Advanced statistical modeling:** Multiple imputation of missing or non-specific data, spatial or spatiotemporal regression modeling, and empirical or hierarchical Bayesian approaches (e.g., CODEm or CoDcorrect analytical algorithms)[11,19,31].
    * **Data integration (record linkage):** Deterministic or probabilistic algorithms linking mortality records with complementary databases (e.g., hospital discharge records, automated verbal autopsies, or epidemiological surveillance systems)[23,26].
    * **Artificial intelligence and machine learning:** Data-driven predictive architectures (e.g., decision trees, gradient boosting, random forests, or other data-mining frameworks) applied to classify or predict underlying causes of death masked by non-specific codes.

- **Exclusion criteria:**
  - Studies analyzing mortality data quality in a purely descriptive manner (e.g., reporting isolated completeness percentages or garbage code prevalences) without implementing or evaluating an active mathematical or algorithmic correction method.
  - Research focused solely on correcting numerical under-registration (omission of death capture), unless the methodology explicitly integrates cause-of-death redistribution.
  - Studies utilizing machine learning, decision trees, or advanced statistical models exclusively for patient-level risk prediction, therapeutic evaluation, or prognostic forecasting (e.g., predicting survival rates following clinical interventions) without the explicit aim of reclassifying, correcting, or redistributing ill-defined causes of death ("garbage codes") in demographic datasets.

2.3 Context

To ensure a comprehensive mapping of the state of the art without geographic selection bias, this review adopts a **global context without economic restrictions**, encompassing mortality reporting systems across both High-Income Countries (HICs) and Low- and Middle-Income Countries (LMICs).

The operational boundaries for this component are structured as follows:

- **Inclusion criteria:**
  - Research situated in any mortality reporting setting—ranging from highly developed Civil Registration and Vital Statistics (CRVS) systems to data-scarce, weak, or fragmented health information systems characterized by low medical certification coverage, high regional heterogeneity, or small-area spatiotemporal misalignments.
  - Studies evaluating, implementing, or discussing the impact of garbage code correction techniques on individual-level microdata disaggregation, the generation of consistent baseline mortality risk matrices, individual trajectory estimations, or parameterizing inputs for health microsimulation platforms (e.g., stochastic microsimulations or agent-based models).
  - Studies originally operating at a macro or aggregate population level, provided they detail the mathematical assumptions, record-level classification algorithms, or linkage protocols necessary to calibrate and parameterize individual-level simulation environments.
- **Exclusion criteria:**
  - Methodologies operating exclusively at an aggregate population level whose mathematical formulations or data structures preclude individual-level micro-disaggregation, thereby lacking utility or transferability for agent-based or individual-level simulation parameterization.

2.4 Sources of evidence

- **Publication types:** Peer-reviewed original research articles published in indexed journals, methodological book chapters, and authoritative institutional gray literature (e.g., technical reports from the WHO, PAHO, or IHME).
- **Timeframe:** Material published from 1996 onwards, capturing the global introduction and widespread adoption of the Tenth Revision of the International Classification of Diseases (ICD-10), which consolidated the term "garbage codes."
- **Languages:** Studies published in English, Spanish, and Portuguese, reflecting the substantial volume of methodological literature generated in Latin America (particularly Brazil).

3 METHODS

3.1 Search strategy

The search strategy will be designed to capture the entire historical continuum of the scientific literature concerning mortality data correction. By encompassing the complete available timeframe, the search guarantees the retrieval of both contemporary machine learning frameworks and the foundational demographic methodologies used to reclassify ill-defined causes of death across all iterations of the International Classification of Diseases.

- **Academic Literature:** To identify academic literature, a three-step search strategy will be developed in collaboration with a medical librarian:
  1. **Initial limited search:** An initial search of MEDLINE (Pubmed) will be undertaken to analyze text words in titles and abstracts, Medical Subject Headings (MeSH), and author-supplied keywords from relevant articles. (Appendix I)
  2. **Full search expansion:** A comprehensive search strategy will be developed using these identified terms, then translated and adapted for Scopus, Web of Science, LILACS, and SciELO. The complete search syntax for MEDLINE (Pubmed) is presented in Appendix I.
  3. **Reference hand-searching:** The reference lists of all selected articles and relevant systematic or scoping reviews will be hand-searched to identify primary sources missed by the electronic search.

- **Grey literature:** To capture unindexed institutional reports, technical documents, and policy papers while ensuring full methodological transparency, grey literature retrieval will follow the JBI guidelines for evidence syntheses.

    1. The search will encompass targeted repository exploration of key international and regional health authorities—specifically the World Health Organization (WHO), the Pan American Health Organization (PAHO), the Institute for Health Metrics and Evaluation (IHME), and the Ministry of Health of El Salvador (MINSAL).
    2. Specialized grey literature databases (e.g., CADTH GreyMatters, BASE, and OpenGrey) will be interrogated. For open web searches conducted via Google Scholar, bulk record management will be systematized using the 'Publish or Perish' software.
    3. In strict adherence to PRISMA-S standards, every targeted website check, exact search string, date of execution, and precise hit count will be systematically logged in an audit tracking table (Appendix II) to guarantee complete auditability and reproducibility.

3.2 Data management and search automation

To ensure the methodological reproducibility and transparency required by the PRISMA-ScR guidelines, this scoping review protocol was registered a priori on the Open Science Framework (OSF) [osf.io/374tm]. The information retrieval process will be systematized within the R statistical environment (v4.5 or higher). Specifically, the `litsearchr` package will be employed for the algorithmic optimization of Boolean syntax via term co-occurrence networks, while indexed databases will be interrogated programmatically through Application APIs using rentrez and rscopus. Data consolidation and duplicate removal will be managed via synthesisr using fuzzy string-matching models, supplemented by manual verification of borderline cases. Title and abstract screening will be conducted via revtools, supported by Latent Dirichlet Allocation (LDA) topic modeling strictly as a computational decision-support tool to reduce reviewer fatigue. To eliminate any risk of algorithmic bias, two reviewers will independently make every inclusion decision, resolving discrepancies through consensus or third-party arbitration.

The final literature search will be re-run immediately prior to submission. Any methodological deviations from this registered protocol will be fully documented and reported in the final publication.

3.3 Data extraction and charting

To ensure integrity, traceability, and reproducibility, we will conduct data extraction programmatically within the R statistical environment (v4.5 or higher). This automated approach mitigates the transcription errors inherent in manual processes and ensures a standardized workflow. We have designed an electronic data charting form to capture information across six dimensions:

- **Study metadata:** Author, year of publication, country, and study design.
- **Geographic and health system context:** Challenges encountered during implementation, such as inadequate medical training or certification capacity, and recommendations for systemic improvement[7].
- **Mortality database characteristics:** Specific baseline registry quality metrics used to evaluate the initial validity of the input data. A prime example is the five-star rating system, which objectively scores mortality data based on demographic completeness and the prevalence of major garbage codes[11], as well as the algorithmic assessment of missing or unexpected values in core demographic fields, which serves as a critical proxy for systemic data quality issues in highly deprived municipalities[37].
- **"Garbage code" typology:** The specific definition of a garbage code applied by the authors[17], alongside the age- and sex-specific epidemiological profiles of these codes, to understand the clinical nature of the missing data at different life stages[11].

- **Correction methodologies:** The type of redistribution model used, the algorithmic and software ecosystems utilized, and the specific performance metrics reported to demonstrate data quality improvement[42].
- **Microsimulation applicability:** The explicit capacity of the methodology to generate highly disaggregated microdata and handle parameter uncertainty for stochastic predictive modeling[43].

This form will be piloted with a random sample of three included studies to refine consistency and ensure variable granularity. Two reviewers will perform the extraction independently, resolving discrepancies through consensus. The resulting data will be consolidated into a relational structure (e.g., tibble or data.frame) to facilitate transparent downstream analysis. The full extraction schema (exported as JSON/CSV) and corresponding R scripts will be made available via the Open Science Framework (OSF) upon study completion. In accordance with Joanna Briggs Institute (JBI) methodology for scoping reviews, a formal critical appraisal of included sources of evidence will not be conducted, as the objective of this review is to map and synthesize the breadth of existing quantitative methodologies rather than to evaluate the internal validity of individual studies. For a detailed breakdown of the variables and the classification framework utilized, refer to the Data Charting Instrument (Appendix IV).

3.4 Data analysis, synthesis, and presentation

This scoping review adheres to the Joanna Briggs Institute methodology. We will conduct all data processing and visualization programmatically within the R statistical environment.

- **Descriptive quantitative analysis:** We will calculate absolute frequencies and percentages to characterize the literature by chronological distribution, geographic reach, taxonomic frameworks, cause categories, and analytical techniques. These analytical techniques will be categorized by their computational complexity, ranging from foundational deterministic methods to advanced predictive modeling, machine learning, and spatial analysis.
- **Thematic narrative synthesis:** We will accompany the extracted data with an explanatory narrative structured around three core analytical axes:
  - **Status and burden of "garbage codes" across social deprivation strata":** Critically describing the prevalence and typology of ill-defined mortality causes in data-scarce settings, specifically analyzing how uncorrected garbage codes disproportionately mask the true mortality burden in populations with higher social deprivation[46], and evaluating the risks of inappropriate centralization when applying macro-level regression algorithms to localized municipal datasets.[47].
  - **Methodological Diversity and Assumptions:** Evaluating the mathematical complexity, infrastructure requirements, and parameter uncertainty handling across algorithmic families.
  - **Transferability to microsimulation and disease burden modeling:** Appraising whether identified redistribution algorithms can be rigorously integrated into agent-based platforms to preserve microdata validity and baseline risk calibration. Drawing upon methodological precedents from prior comparative risk assessments in Latin America—such as the 2020 baseline burden estimations of sugar-sweetened beverages in El Salvador[48]—the synthesis will critically evaluate how uncorrected vital statistics distort baseline mortality risks for non-communicable diseases (e.g., type 2 diabetes and ischemic heart disease) across socioeconomically deprived subgroups.
- **Mapping and visualization:** To visualize the structure and knowledge gaps of the field, we will generate advanced graphics using ggplot2[49] and igraph[50]. This will include structured evidence tables cross-referencing correction methods with their input data requirements, two-dimensional heatmaps mapping analytical methods against specific ICD Chapters, and flow network diagrams illustrating the technical pathway required to connect traditional vital statistics with microsimulation parameterization.

4 Ethical approval

As this scoping review relies exclusively on publicly available published scientific literature, official institutional technical reports, and aggregated population-based secondary data (Civil Registration and Vital Statistics systems), formal ethical approval by an Institutional Review Board (IRB) or ethics committee was not required.

5 Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this protocol, the authors used Gemini Notebook to assist in scanning source PDF documents for the data extraction matrix—followed by 100% human verification of all data points—and for language polishing and proofreading. All evidence selection, critical appraisal, methodological synthesis, and conceptual drafting were performed independently by the authors, who assume full responsibility for the integrity and accuracy of the final content.

6 Availability of data and materials

All datasets generated or analyzed during this study are included in this published protocol and its supplementary information files. The complete R scripts, customized search syntaxes, and electronic data charting instruments are publicly accessible via the Open Science Framework (OSF) repository [osf.io/374tm]. Any additional customized scripts, codebooks, or intermediate extraction matrices are available from the corresponding author upon reasonable request.s of Interest.

The authors declare that they have no competing interests. This scoping review was conducted solely for academic, educational, and methodological research purposes, with no commercial, financial, corporate, or industrial support that could be construed as a potential conflict of interest.

7 Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. The study was conducted entirely through the academic time, institutional affiliation resources, and professional dedication of the authors.

8 Authors' contributions

- **Giovanni Guevara-Vásquez**: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – Original Draft, Visualization, Project administration.
- **Pablo Salazar-Colocho**: Conceptualization, Methodology, Writing – Review & Editing.
- **David Tejada-Peña**: Investigation, Resources, Data curation, Writing – Review & Editing.
- **Hazel García-Sanchez**: Investigation, Resources, Data curation, Writing – Review & Editing.
- **César Gavidia-Leiva**: Investigation, Resources, Data curation, Writing – Review & Editing.
- **Mirna Melendez-Gálvez**: Investigation, Resources, Data curation, Writing – Review & Editing.

- **Kevin Siliezar-Madrid**: Methodology (Information Retrieval), Investigation, Data Curation (Search execution in R), Writing – Review & Editing.
- **Alejandra Rivas-Portillo**: Methodology, Investigation (Dual independent screening and selection), Data Curation, Writing – Review & Editing.

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10 Appendices

10.0.1 Appendix I: Search Strategy (PubMed / MEDLINE)

Database: PubMed (via NCBI API) **Date of Search:** October 17, 2026 **Limits:** None (No temporal or language restrictions applied). **Note:** The syntax was optimized using the `litsearchr` package to assemble free-text terms, combined manually with National Library of Medicine controlled vocabulary (`[Mesh]`).

Line	Search Strategy (PubMed Syntax)	Conceptual Block
1	`Vital Statistics[Mesh] OR Death Certificates[Mesh] OR Cause of Death[Mesh] OR Mortality[Mesh]`	**Population (MeSH):** Core mortality registry indexing
2	`("vital statistics" OR "registration of vital statistics" OR "vital registration" OR "civil registration" OR CRVS OR "population register*" OR "death regist*" OR "death certificat*" OR "death record*" OR "cause of death" OR "causes of death" OR "death cause*" OR "underlying cause*" OR "death rate*" OR "population surveillance" OR "public health surveillance" OR "international classification of disease*" OR ICD-9 OR ICD-10 OR ICD-11 OR "clinical coding" OR "medical coding" OR "health information system*" OR mortality)[tiab]`	**Population (Text Words):** Vital statistics and national death registry systems
3	`#1 OR #2`	**Result:** Core Mortality Data

Line	Search Strategy (PubMed Syntax)	Conceptual Block
4	("garbage code*" OR "garbage cause*" OR "garbage categor*" OR "garbage diagnos*" OR "poorly defined cause*" OR "causas mal definidas" OR "codigos basura" OR "codigos lixo" OR "codigos garbage" OR R00-R99 OR ill-defined OR "unknown cause*" OR "unspecified cause*" OR "nonspecific cause*" OR "non-specific cause*" OR "unspecific cause*" OR "undetermined cause*" OR "vague code*" OR "cause of death assignment" OR "underlying cause assignment" OR misclassif* OR miscod* OR misattribut* OR miscertif*)[tiab]	**Concept A (Text Words):** Specific typology of data errors ("Garbage Codes")
5	Models, Statistical[Mesh] OR Data Interpretation, Statistical[Mesh] OR Bayes Theorem[Mesh] OR Spatio-Temporal Analysis[Mesh] OR Medical Record Linkage[Mesh] OR Autopsy[Mesh] OR Patient Discharge[Mesh] OR Artificial Intelligence[Mesh] OR Decision Trees[Mesh] OR Data Mining[Mesh] OR Random Forest[Mesh] OR Life Tables[Mesh] OR Demography[Mesh] OR Algorithms[Mesh] OR Computer Simulation[Mesh]	**Concept B (MeSH):** Indexing for analytical and computational methods

Line	Search Strategy (PubMed Syntax)	Conceptual Block
6	("proportional redistribution" OR "proportional distribution" OR "fractional algorithm*" OR "Brass technique" OR "Brass technic" OR "reverse survival" OR "stable population" OR "demographic analys*" OR "life table*" OR "statistical model*" OR "probabilistic model*" OR bayesian OR "empirical bayes" OR spatiotemporal OR spatio-temporal OR "spatial regression" OR imput* OR CODEm OR CoDCorrect OR "cause of death ensemble model*" OR "coarsened exact matching" OR reallocat* OR reassign* OR recod* OR algorithm* OR "multiple cause*" OR linkage OR "verbal autops*" OR "hospital discharge" OR "discharge record*" OR "discharge data" OR "artificial intelligence" OR "machine learning" OR "decision tree*" OR "data mining" OR "random forest*" OR "gradient boosting" OR "deep learning" OR "neural network*" OR microsimulat* OR micro-simulation* OR agent-based OR "synthetic population*" OR "global burden of disease" OR GBD OR InterVA OR InSilicoVA OR SmartVA OR "tariff method" OR "bridge coding" OR "comparability ratio*" OR "IRIS software" OR "automated coding" OR "automated classification" OR "small area estimation" OR "correction factor*" OR redistribut* OR reclassif*) [tiab]	**Concept B (Text Words):** Specific statistical, computational, and redistribution frameworks

Line	Search Strategy (PubMed Syntax)	Conceptual Block
7	`#5 OR #6`	**Result:** Methodological Ecosystem
8	`#3 AND #4 AND #7`	**FINAL COMBINED STRATEGY**

10.0.1.1 Programmatic Search Execution (R Environment) In adherence to PRISMA-S guidelines for computational reproducibility, the final search query will be executed programmatically via the NCBI API using the `rentrez` package. The exact search string assembled for the programmatic query is documented below:

```
# Full query assembly for rentrez implementation
query_pubmed <- paste0(
  "((Vital Statistics[Mesh] OR Death Certificates[Mesh] OR Cause of Death[Mesh] OR Mortality[Mesh]) OR
  '("vital statistics" OR "registration of vital statistics" OR "vital registration" OR "civil registra
  " AND ",
  '("garbage code*" OR "garbage cause*" OR "garbage categor*" OR "garbage diagnos*" OR "poorly defined
  " AND ",
  "((Models, Statistical[Mesh] OR Data Interpretation, Statistical[Mesh] OR Bayes Theorem[Mesh] OR Spati
  '("proportional redistribution" OR "proportional distribution" OR "fractional algorithm*" OR "Brass t
)
```

10.0.2 Search Strategy: Scopus (Elsevier)

Database: Scopus **Date of Search:** October 17, 2026 **Limits:** None (No publication date or language restrictions applied). **Note:** The search syntax was adapted for Scopus using the `TITLE-ABS-KEY` field code to simultaneously search across titles, abstracts, and keywords. Due to strict API character and boolean operator constraints for highly complex queries, this search is executed directly via the Scopus Advanced Search web interface. The retrieved records are exported in `.ris` format for programmatic deduplication via the `synthesisr` package in R.

Line	Search Strategy (Scopus Syntax)	Conceptual Block
1	`TITLE-ABS-KEY("vital statistics" OR "registration of vital statistics" OR "vital registration" OR "civil registration" OR CRVS OR "population register*" OR "death regist*" OR "death certificat*" OR "death record*" OR "cause of death" OR "causes of death" OR "death cause*" OR "underlying cause*" OR "death rate*" OR "population surveillance" OR "public health surveillance" OR "international classification of disease*" OR ICD-9 OR ICD-10 OR ICD-11 OR "clinical coding" OR "medical coding" OR "health information system*" OR mortality)`	**Population:** Vital statistics and national death registry systems
2	`TITLE-ABS-KEY("garbage code*" OR "garbage cause*" OR "garbage categor*" OR "garbage diagnos*" OR "poorly defined cause*" OR "causas mal definidas" OR "codigos basura" OR "codigos lixo" OR "codigos garbage" OR R00-R99 OR ill-defined OR "unknown cause*" OR "unspecified cause*" OR "nonspecific cause*" OR "non-specific cause*" OR "unspecific cause*" OR "undetermined cause*" OR "vague code*" OR "cause of death assignment" OR "underlying cause assignment" OR misclassif* OR miscod* OR misattribut* OR miscertif*)`	**Concept A:** Specific typology of data errors ("Garbage Codes")

Line	Search Strategy (Scopus Syntax)	Conceptual Block
3	TITLE-ABS-KEY("statistical model*" OR "statistical data interpretation" OR "Bayes theorem" OR "spatiotemporal analysis" OR "medical record linkage" OR autopsy OR "patient discharge" OR "artificial intelligence" OR "decision tree*" OR "data mining" OR "random forest*" OR "life table*" OR demography OR algorithm* OR "computer simulation" OR "proportional redistribution" OR "proportional distribution" OR "fractional algorithm*" OR "Brass technique" OR "Brass technic" OR "reverse survival" OR "stable population" OR "demographic analys*" OR "probabilistic model*" OR bayesian OR "empirical bayes" OR spatiotemporal OR spatio-temporal OR "spatial regression" OR imput* OR CODEm OR CoDCorrect OR "cause of death ensemble model*" OR "coarsened exact matching" OR reallocat* OR reassign* OR recod* OR "multiple cause*" OR linkage OR "verbal autops*" OR "hospital discharge" OR "discharge record*" OR "discharge data" OR "machine learning" OR "gradient boosting" OR "deep learning" OR "neural network*" OR microsimulat* OR micro-simulation* OR agent-based OR "synthetic population*" OR "global burden of disease" OR GBD OR InterVA OR InSilicoVA OR SmartVA OR "tariff method" OR "bridge coding" OR "comparability ratio*" OR "IRIS software" OR "automated coding" OR "automated	**Concept B:** Analytical, statistical, and computational methods

Line	Search Strategy (Scopus Syntax)	Conceptual Block
4	`#1 AND #2 AND #3`	**FINAL COMBINED STRATEGY**

10.0.2.1 Web Interface Execution String The following complete syntax was assembled to be copied and pasted directly into the Scopus Advanced Search query box:

"'text TITLE-ABS-KEY("vital statistics" OR "registration of vital statistics" OR "vital registration" OR "civil registration" OR CRVS OR "population register*" OR "death regist*" OR "death certificat*" OR "death record*" OR "cause of death" OR "causes of death" OR "death cause*" OR "underlying cause*" OR "death rate*" OR "population surveillance" OR "public health surveillance" OR "international classification of disease*" OR ICD-9 OR ICD-10 OR ICD-11 OR "clinical coding" OR "medical coding" OR "health information system*" OR mortality) AND TITLE-ABS-KEY("garbage code*" OR "garbage cause*" OR "garbage categor*" OR "garbage diagnos*" OR "poorly defined cause*" OR "causas mal definidas" OR "codigos basura" OR "codigos lixo" OR "codigos garbage" OR R00-R99 OR ill-defined OR "unknown cause*" OR "unspecified cause*" OR "nonspecific cause*" OR "non-specific cause*" OR "unspecific cause*" OR "undetermined cause*" OR "vague code*" OR "cause of death assignment" OR "underlying cause assignment" OR misclassif* OR miscod* OR misattribut* OR miscertif) AND TITLE-ABS-KEY("statistical model*" OR "statistical data interpretation" OR "Bayes theorem" OR "spatiotemporal analysis" OR "medical record linkage" OR autopsy OR "patient discharge" OR "artificial intelligence" OR "decision tree*" OR "data mining" OR "random forest*" OR "life table*" OR demography OR algorithm* OR"computer simulation" OR "proportional redistribution" OR "proportional distribution" OR "fractional algorithm*" OR "Brass technique" OR "Brass technic" OR "reverse survival" OR "stable population" OR "demographic analys*" OR "probabilistic model*" OR bayesian OR "empirical bayes" OR spatiotemporal OR spatio-temporal OR "spatial regression" OR imput* OR CODEm OR CoDCorrect OR"cause of death ensemble model*" OR "coarsened exact matching" OR reallocat* OR reassign* OR recod* OR "multiple cause*" OR linkage OR "verbal autops*" OR "hospital discharge" OR "discharge record*" OR "discharge data" OR "machine learning" OR "gradient boosting" OR "deep learning" OR "neural network*" OR microsimulat* OR micro-simulation* OR agent-based OR "synthetic population*" OR "global burden of disease" OR GBD OR InterVA OR InSilicoVA OR SmartVA OR "tariff method" OR "bridge coding" OR "comparability ratio*" OR "IRIS software" OR "automated coding" OR "automated classification" OR "small area estimation" OR "correction factor*" OR redistribut* OR reclassif*)

10.0.3 Search Strategy: Web of Science (Core Collection)

Database: Web of Science (Clarivate Analytics) **Date of Search:** October 17, 2026 **Limits:** None (No publication date or language restrictions applied). **Note:** The search syntax was adapted for Web of Science using the `TS=` (Topic) field tag, which queries the Title, Abstract, Author Keywords, and Keywords Plus fields. Due to the high complexity and length of the Boolean string, the search is executed directly via the Web of Science Advanced Search interface. Retrieved records are exported in `.ris` format for downstream integration.

Line	Search Strategy (Web of Science Syntax)	Conceptual Block
1	TS=("vital statistics" OR "registration of vital statistics" OR "vital registration" OR "civil registration" OR CRVS OR "population register*" OR "death regist*" OR "death certificat*" OR "death record*" OR "cause of death" OR "causes of death" OR "death cause*" OR "underlying cause*" OR "death rate*" OR "population surveillance" OR "public health surveillance" OR "international classification of disease*" OR ICD-9 OR ICD-10 OR ICD-11 OR "clinical coding" OR "medical coding" OR "health information system*" OR mortality)	**Population:** Vital statistics and national death registry systems
2	TS=("garbage code*" OR "garbage cause*" OR "garbage categor*" OR "garbage diagnos*" OR "poorly defined cause*" OR "causas mal definidas" OR "codigos basura" OR "codigos lixo" OR "codigos garbage" OR R00-R99 OR ill-defined OR "unknown cause*" OR "unspecified cause*" OR "nonspecific cause*" OR "non-specific cause*" OR "unspecific cause*" OR "undetermined cause*" OR "vague code*" OR "cause of death assignment" OR "underlying cause assignment" OR misclassif* OR miscod* OR misattribut* OR miscertif*)	**Concept A:** Specific typology of data errors ("Garbage Codes")

Line	Search Strategy (Web of Science Syntax)	Conceptual Block
3	TS=("statistical model*" OR "statistical data interpretation" OR "Bayes theorem" OR "spatiotemporal analysis" OR "medical record linkage" OR autopsy OR "patient discharge" OR "artificial intelligence" OR "decision tree*" OR "data mining" OR "random forest*" OR "life table*" OR demography OR algorithm* OR "computer simulation" OR "proportional redistribution" OR "proportional distribution" OR "fractional algorithm*" OR "Brass technique" OR "Brass technic" OR "reverse survival" OR "stable population" OR "demographic analys*" OR "probabilistic model*" OR bayesian OR "empirical bayes" OR spatiotemporal OR spatio-temporal OR "spatial regression" OR imput* OR CODEm OR CoDCorrect OR "cause of death ensemble model*" OR "coarsened exact matching" OR reallocat* OR reassign* OR recod* OR "multiple cause*" OR linkage OR "verbal autops*" OR "hospital discharge" OR "discharge record*" OR "discharge data" OR "machine learning" OR "gradient boosting" OR "deep learning" OR "neural network*" OR microsimulat* OR micro-simulation* OR agent-based OR "synthetic population*" OR "global burden of disease" OR GBD OR InterVA OR InSilicoVA OR SmartVA OR "tariff method" OR "bridge coding" OR "comparability ratio*" OR "IRIS software" OR "automated coding" OR "automated	**Concept B:** Analytical, statistical, and computational methods

Line	Search Strategy (Web of Science Syntax)	Conceptual Block
4	`#1 AND #2 AND #3`	**FINAL COMBINED STRATEGY**

```
TS=("vital statistics" OR "registration of vital statistics" OR "vital registration" OR "civil registra
```

10.0.3.1 Web Interface Execution String

10.0.4 Search Strategy: LILACS (Latin American and Caribbean Health Sciences Literature)

Database: LILACS (via Virtual Health Library - VHL / BVS) **Date of Search:** October 17, 2026 **Limits:** None (No publication date restrictions applied). **Note:** The search syntax for LILACS required methodological adaptation due to the character limits and Boolean nesting constraints of the VHL (BVS) search interface. The highly complex string used for Scopus and Web of Science was distilled into its core conceptual components. Furthermore, to maximize sensitivity within regional literature, free-text terms were translated into a trilingual format (English, Spanish, and Portuguese) mapping to DeCS (Health Sciences Descriptors) terminology. The search is executed using the `tw:` (text word) field code.

Line	Search Strategy (LILACS / VHL Syntax)	Conceptual Block
1	`tw:("vital statistics" OR "estadisticas vitales" OR "estatisticas vitais" OR "cause of death" OR "causas de muerte" OR "causas de morte" OR "death certificate*" OR "certificados de defuncion" OR "declaracao de obito" OR CRVS OR mortality OR mortalidad OR mortalidade)`	**Population:** Mortality registries and vital statistics (Trilingual)
2	`tw:("garbage code*" OR "codigos basura" OR "codigos lixo" OR "codigos garbage" OR "ill-defined" OR "mal definidas" OR "mal definidos" OR "R00-R99" OR "unspecified cause*" OR "causas no especificadas" OR "causas nao especificadas")`	**Concept A:** "Garbage Codes" and ill-defined causes (Trilingual)

Line	Search Strategy (LILACS / VHL Syntax)	Conceptual Block
3	`tw:(redistribut* OR redistribucion OR redistribuicao OR reclassif* OR reclasificacion OR reclassificacao OR imput* OR algorithm* OR algoritmo* OR "machine learning" OR "aprendizaje automatico" OR "aprendizado de maquina" OR "artificial intelligence" OR "inteligencia artificial" OR "proportional distribution" OR "distribucion proporcional" OR "multiple cause*" OR "causas multiples" OR "causas multiplas")`	**Concept B:** Analytical and redistribution methods (Trilingual)
4	`#1 AND #2 AND #3`	**FINAL COMBINED STRATEGY**

```
(tw:("vital statistics" OR "estadisticas vitales" OR "estatisticas vitais" OR "cause of death" OR "causa
```

10.0.4.1 Web Interface Execution String

10.0.5 Search Strategy: SciELO (Scientific Electronic Library Online)

Database: SciELO **Date of Search:** October 17, 2026 **Limits:** None (No publication date restrictions applied). **Note:** To effectively capture the substantial volume of methodological literature generated in Latin America, particularly from Brazil regarding "códigos lixo" (garbage codes), the SciELO search strategy was adapted into a targeted, trilingual format (English, Spanish, Portuguese). Due to the Boolean complexity limits of the SciELO search engine, the string was distilled to its core programmatic concepts. The search was executed in the "All indices" default field. Retrieved records are exported in `.ris` format for synthesis.

Line	Search Strategy (SciELO Syntax)	Conceptual Block
1	("vital statistics" OR "estadisticas vitales" OR "estatisticas vitais" OR "cause of death" OR "causas de muerte" OR "causas de morte" OR "mortality" OR "mortalidad" OR "mortalidade" OR "death certificate*" OR "certificados de defuncion" OR "declaracao de obito")	**Population:** Mortality registries and vital statistics (Trilingual)
2	("garbage code*" OR "codigos basura" OR "codigos lixo" OR "codigos garbage" OR "ill-defined" OR "mal definidas" OR "mal definidos" OR "unspecified cause*" OR "causas no especificadas" OR "causas nao especificadas" OR R00-R99)	**Concept A:** "Garbage Codes" and ill-defined causes (Trilingual)
3	(redistribut* OR redistribucion OR redistribuicao OR reclassif* OR reclasificacion OR reclassificacao OR imput* OR algorithm* OR algoritmo* OR "machine learning" OR "aprendizaje automatico" OR "aprendizado de maquina" OR "proportional distribution" OR "distribucion proporcional")	**Concept B:** Analytical and redistribution methods (Trilingual)
4	#1 AND #2 AND #3	**FINAL COMBINED STRATEGY**

10.0.5.1 Web Interface Execution String text ("vital statistics" OR "estadisticas vitales" OR "estatisticas vitais" OR "cause of death" OR "causas de muerte" OR "causas de morte" OR "mortality" OR "mortalidad" OR "mortalidade" OR "death certificate*" OR "certificados de defuncion" OR "declaracao de obito") AND ("garbage code*" OR "codigos basura" OR "codigos lixo" OR "codigos garbage" OR "ill-defined" OR "mal definidas" OR "mal definidos" OR "unspecified cause*" OR "causas no especificadas" OR "causas nao especificadas" OR R00-R99) AND (redistribut* OR redistribucion OR redistribuicao OR

```
reclassif\* OR reclasificacion OR reclassificacao OR imput\* OR algorithm\* OR algoritmo\*
OR "machine learning" OR "aprendizaje automatico" OR "aprendizado de maquina" OR "proportional
distribution" OR "distribucion proporcional")
```

10.1 Appendix II: PRISMA-S audit and mapping table

The following table details the adherence to the 16 items of the PRISMA-S guideline for the scoping review protocol entitled *"Methodologies for the correction of Garbage codes in mortality records and their applicability in microsimulation models"*, ensuring complete search auditability, transparency, and computational reproducibility.

PRISMA-S Item	Reporting Requirement	Specific Implementation in Protocol & R Pipeline	Location / Status
1. Information sources	Describe all information sources consulted (databases, registries, websites, grey literature).	Indexed databases: PubMed/MEDLINE, Scopus, Web of Science, LILACS, and SciELO. Institutional grey literature: WHO, PAHO, IHME, and MINSAL. Specialized grey repositories: CADTH GreyMatters, BASE, and OpenGrey.	Section 3.1 & Section 3.2
2. Electronic search strategy	Present the full electronic search strategy for at least one major database, including syntax and operators.	Complete search query designed for MEDLINE (via Ovid) structured under the Population, Concept, and Context (PCC) framework.	Section 6.1 (Appendix I)
3. Database platforms and vendors	Specify the database platforms or interface vendors used for each search.	Programmatic querying via Application APIs and specialized R libraries (`rentrez` for PubMed, `rscopus` for Scopus) and Ovid interface for MEDLINE.	Section 3.2
4. Translation of search strategy	Describe how the search strategy was adapted for different thesauri and syntaxes.	Adaptation of MeSH descriptors and free-text terms optimized via term co-occurrence networks using the `litsearchr` package in R.	Section 3.2
5. Limits and restrictions	Specify any search-level restrictions applied (dates, languages, document types).	No temporal or publication date restrictions were applied. No initial language restrictions (English, Spanish, Portuguese).	Section 2.4 & Section 6.1

PRISMA-S Item	Reporting Requirement	Specific Implementation in Protocol & R Pipeline	Location / Status
6. Search dates	Indicate the exact date when searches were executed or scheduled for each source.	Primary literature search scheduled *a priori* for **October 2026**, with a final verification re-run prior to synthesis.	Section 3.1 & Section 6.1
7. Contacting authors	Indicate if study authors, experts, or key researchers were contacted to identify unpublished studies.	Planned consultations with vital statistics epidemiologists, public health officials, and key regional collaborators in Latin America.	Section 3.1
8. Registration	Cite the *a priori* protocol registration and registry platform used.	Registered *a priori* on the **Open Science Framework (OSF)**.	Introduction & Section 3.3
9. Supplementary searches (Citation searching)	Describe supplementary search methods (backward and forward citation tracking).	Manual backward citation tracking (*backwards citation searching*) on all studies selected for full-text extraction.	Section 3.1
10. Peer review of search strategy	Indicate if the search strategy was peer-reviewed by an expert (e.g., librarian or PRESS).	Search strategy developed in collaboration with a medical librarian and algorithmically optimized via term co-occurrence networks in R.	Section 3.1 & Section 3.2
11. Total records identified	Report the total number of records identified from each database and grey literature source.	*Pending empirical execution (October 2026).* Will be formally documented in the final PRISMA-ScR flow diagram.	Flow Diagram / Results
12. Duplicate removal	Describe the process and tools used for duplicate removal.	Automated consolidation using the `synthesisr` package in R via fuzzy string-matching, with manual verification of edge cases.	Section 3.2
13. Selection process	Describe the study selection and screening process (title/abstract and full-text screening).	Dual independent screening by two human reviewers via `revtools`, assisted by Latent Dirichlet Allocation (LDA) topic modeling strictly as a human-in-the-loop decision-support tool.	Section 3.2

PRISMA-S Item	Reporting Requirement	Specific Implementation in Protocol & R Pipeline	Location / Status
14. Data collection process	Describe the data collection process and extraction tools.	Standardized programmatic data charting in R across six analytical dimensions, executed independently and in duplicate.	Section 3.3
15. Data items	List and define all variables planned for extraction.	Electronic data charting form structured across 6 blocks: Metadata, Health System Context, Registry Characteristics, Garbage Code Typology, Correction Methodology, and Microsimulation Applicability.	Section 6.2 (Appendix II)
16. Search auditability & reproducibility	Ensure complete audit trail and reproducibility of the search process.	Executable R scripts, syntax codes, and extraction matrices exported and permanently archived in the project's OSF repository.	Section 3.2 & Section 3.3

10.2 Appendix III: Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) Checklist

Section / Topic	Item #	PRISMA-ScR Checklist Item	Reported on Page / Section	Specific Implementation in Protocol
TITLE				
Title	1	Identify the report as a scoping review.	Title Page & Section 1	Declared as a Scoping Review Protocol in the main title.
ABSTRACT				
Structured summary	2	Provide a structured summary including: background, objectives, eligibility criteria, sources of evidence, charting methods, and main synthesis plan.	Section 1 (Abstract)	Structured abstract following JBI and PRISMA-ScR standards.
INTRODUCTION				

Section / Topic	Item #	PRISMA-ScR Checklist Item	Reported on Page / Section	Specific Implementation in Protocol
Rationale	3	Describe the rationale for the review in the context of what is already known.	Section 1.1 (Background)	Details the "garbage codes" problem, limitations of macro-level corrections, and the translational gap toward microsimulation.
Objectives	4	Provide an explicit statement of the question(s) or objective(s) being addressed.	Section 1.2 (Research Questions)	Formulates 4 explicit research questions using the Population, Concept, and Context (PCC) framework.
METHODS				
Protocol & registration	5	Indicate if a review protocol exists; state if registered and supply registration info.	Section 3.2 & OSF Link	Registered *a priori* on the Open Science Framework (OSF ID: `10.17605/OSF.IO/ZP9AQ`).
Eligibility criteria	6	Specify characteristics used to decide eligibility (e.g., PCC, study design).	Section 2 (Eligibility Criteria)	Structured according to Population (mortality records/LMICs), Concept (garbage code algorithms), and Context (microsimulation applicability).
Information sources	7	Describe all information sources (databases, grey literature, contact with authors).	Section 3.1 & Appendix I/III	MEDLINE/Ovid, Scopus, Web of Science, LILACS, SciELO, plus WHO, PAHO, IHME grey literature repositories.
Search strategy	8	Present full electronic search strategy for at least one database, including limits.	Appendix I (Section 7.1)	Complete Ovid MEDLINE syntax provided with Boolean operators, MeSH terms, and line-by-line targets.

Section / Topic	Item #	PRISMA-ScR Checklist Item	Reported on Page / Section	Specific Implementation in Protocol
Selection process	9	State process for selecting sources (screening, eligibility, deduplication tools).	Section 3.2	Programmatic deduplication (`synthesisr`), dual independent screening (`revtools`), assisted by LDA topic modeling.
Data charting process	10	Describe methods of charting data (forms, piloting, reviewer duplication).	Section 3.3 & Appendix II	Dual independent programmatic charting using a 6-block relational data extraction schema in R.
Data items	11	List and define all variables for which data were sought.	Section 3.3 & Appendix II	Defines 20 variables across metadata, health context, registry quality, garbage code typology, methods, and TRL/microsimulation applicability.
Critical appraisal	12	State whether critical appraisal was conducted; if not, state why.	Section 3.3	In accordance with JBI scoping review guidelines, formal risk-of-bias/critical appraisal is omitted as the objective is to map methodologies.
Synthesis of results	13	Describe methods of handling and summarizing data (tables, charts, narrative).	Section 3.4 & 3.5	Combines descriptive quantitative metrics, thematic narrative synthesis, ggplot2 matrix heatmaps, and igraph network diagrams.
RESULTS				
Selection of sources	14	Report numbers of records screened, assessed for eligibility, and included.	*Pending Execution*	To be reported in the PRISMA-ScR Flow Diagram upon study completion (September 2026).

Section / Topic	Item #	PRISMA-ScR Checklist Item	Reported on Page / Section	Specific Implementation in Protocol
Characteristics of sources	15	Describe characteristics of included sources of evidence.	*Pending Execution*	To be tabulated in the final review using the Appendix II extraction schema.
Critical appraisal results	16	Share results of any critical appraisal of individual sources.	*Not Applicable*	Not conducted, in strict adherence to JBI scoping review methodology.
Results of individual sources	17	For each source, present relevant data charted.	*Pending Execution*	Will be archived as open CSV/JSON matrices on the OSF project repository.
Synthesis of results	18	Summarize and/or present the charting results in relation to objectives.	*Pending Execution*	Will map analytical techniques against microsimulation parameterization potential.
DISCUSSION				
Summary of evidence	19	Summarize main findings; link to objectives and broader literature.	*Pending Execution*	To be drafted upon synthesis completion.
Limitations	20	Discuss limitations of the scoping review process.	*Pending Execution*	To be discussed in the final review report.
Conclusions	21	Provide explicit conclusions linked to objectives and implications.	*Pending Execution*	To be provided in the final manuscript.
FUNDING				
Funding	22	Describe sources of funding and role of funders.	Section 7	Self-funded through institutional academic time; no commercial or external grants.

10.3 Appendix IV: Data extraction instrument

Block	Variable	Description / Purpose	Suggested Format / Values
I. Identification & Metadata	Study ID	Unique identifier assigned by R workflow	Numeric (e.g., 001)
	Bibliographic Data	Author, year, title, and journal/institution	Text

Block	Variable	Description / Purpose	Suggested Format / Values
	Evidence Type	Original article, technical report, or method book	Dropdown
II. Geographic & Health Context	Geographic Scope	Country, region, or sub-national area	Text
	Economic Classification	World Bank income level	Dropdown (Low, Lower-Middle, Upper-Middle)
	CRVS System Status	Description of Civil Registration and Vital Statistics status	Text
III. Mortality Database	Data Periodicity	Years covered by mortality records	Text / Numeric
	Data Volume	Sample size (number of deaths analyzed)	Numeric
	Primary Source	Origin of data (e.g., forensic, hospital, surveys)	Text
	ICD Framework	Version of the ICD utilized (ICD-9, ICD-10, ICD-11)	Dropdown
IV. Garbage Code Typology	Definition Framework	Criteria used to identify "garbage" (e.g., GBD list)	Text
	Evaluated ICD Codes	Specific codes being targeted for correction	Text
	Baseline Magnitude	Percentage/volume of garbage codes in original records	Percentage (%)
V. Correction Methodology	Central Technique	Algorithmic classification	Dropdown (e.g., MICE, Bayesian, ML)
	Theoretical Assumptions	Underlying mathematical principles/assumptions	Text
	Software Ecosystem	Tools used (R, SAS, Stata, etc.)	Text
	Uncertainty/Bias Report	Reported confidence intervals or limitations	Yes / No / Partial
VI. Microsimulation Applicability	TRL (Technology Readiness Level)	Maturity of the development method (1-4)	Scale (1-4)
	Output Format	Aggregate vs. individual-level data output	Dropdown
	Demographic Granularity	Consistency at sub-group/small-area levels	Scale (1-5)
	Parametrization Potential	Capability to feed agent-based/microsimulation models	Scale (1-5)

Block	Variable	Description / Purpose	Suggested Format / Values
	Reviewer Notes	Professional notes on applicability and transferability	Free text

10.4

10.5